

Mahim Rawal

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EDUCATION

TU, Khwopa College of Engineering

Bhaktapur, Nepal

Bachelor of Computer Engineering

2021 - 2025

- **Relevant Coursework:** Object-Oriented Programming, Data Mining, Big Data, Database Management Systems, Data Structures & Algorithms, Artificial Intelligence , Computer Network and Securities

PROJECTS

Video Hosting Backend API | *Express, MongoDB, Cloudinary*

- Engineered a scalable RESTful API using Node.js, Express, and MongoDB, utilizing aggregation pipelines to track user watch history.
- Implemented secure, stateless authentication using JWT (access & refresh tokens) and middleware to protect sensitive endpoints.
- Integrated Cloudinary SDK for efficient cloud-based video storage, ensuring optimized media upload and retrieval speeds.

Dynamic Blog Platform | *Express, MongoDB, EJS*

- Developed a server-side rendered (SSR) application using Node.js and Express, utilizing EJS templating for dynamic content generation.
- Architected a MongoDB database with Mongoose schemas to handle data persistence and input validation for articles.
- Implemented full CRUD functionality and RESTful routing to manage blog posts effectively.

Lip Reading Using CNN | *Python , Pytorch , Streamlit*

- Developed a hybrid Deep Learning model combining 3D CNNs for spatial feature extraction and BiLSTMs for temporal sequence analysis
- Developed an interactive web interface using Streamlit, allowing users to upload custom video files for real-time speech decoding.
- Published research paper on this project: [View Publication](#)

Detection of Parkinson's Disease Using Deep Learning | *Python , Pytorch*

- Engineered a diagnostic system analyzing both handwriting images and voice recordings for early disease detection.
- Implemented VGG-16 and LeNet-5 CNN architectures to accurately classify tremor patterns in spiral drawings.
- Applied Random Forest, SVM, and Decision Trees to process audio data and identify vocal impairments.

To-do list | *JavaScript, Html ,CSS*

- Designed a logic-driven frontend using Vanilla JavaScript, enabling users to manage tasks with specific due dates and priority levels.
- Implemented conditional rendering logic to dynamically assign color-coded visual indicators (CSS classes) based on task urgency.
- Utilized LocalStorage to persist complex task objects (including date and priority metadata) across browser sessions.

TECHNICAL SKILLS

Programming Languages: JavaScript, HTML, CSS, SQL , NoSQL, Python (Pandas, NumPy),

Data & Analytics: MongoDB, MySQL, PostgreSQL

Frameworks/Tools: Express.js, Mongoose, Streamlit , TensorFlow, OpenCV

Specializations: Backend Development, Database Management ,Scalable RESTful APIs , Machine Learning Algorithms

CERTIFICATIONS

Oracle Cloud Infrastructure (OCI) Generative AI Professional